



Joseph Penlap Tamagoua

Researcher – Mathematical Modeller

[MyWebsite](#)

Born 27th May 2000

Cameroonian

(+33) 666245954

josephpenlap2000@gmail.com

Education

Université Côte d'Azur

PhD in Applied mathematics

Oct. 2022 - Feb. 2026

Nice, France

- Title: Ecophysiological modelling of plant–nematode interactions: Understanding the origins and consequences of differential plant susceptibility [\[pdf\]](#)
- Advisors: Frédéric Grognard, Valentina Baldazzi, Suzanne Touzeau

African Institute for Mathematical Sciences (AIMS)

Master in Mathematical Sciences, Data Science

2021 - 2022

Limbe, Cameroon

- Title: Modelling the dynamics of a forest environment: role of water cycle
- Advisor: Nathalie Verdière

University of Dschang

Master in Mathematics, Analysis

2019 - 2021

Dschang, Cameroon

- Title: Spaces of locally uniformly bounded functions and applications
- Advisor: Jean Louis Woukeng

University of Dschang

Bachelor in Mathematics and Computer Science

2016 - 2019

Dschang, Cameroon

Work Experience

Temporary Teaching and Research Assistant (ATER)

Polytech Nice Sophia

Sept. 2025 - Aug. 2026

Sophia Antipolis, France

Research visit - [Garett's lab](#)

University of Florida - Plant pathology department

Dec. 2023

Florida, USA

Doctoral Researcher - *Inria-INRAE grant*

Inria, [Macbes](#) team & INRAE, Institut Sophia Agrobiotech / [M2P2](#)

Oct. 2022 - Sept. 2025

Sophia Antipolis, France.

Research Intern

Mathematics Research Unit (URMA)

Apr. 2019 - July 2021

Dschang, Cameroon

Teaching Experience

Lecturer

Polytech Nice Sophia

Sept. 2025 - Aug. 2026

Sophia Antipolis, France

- Linear Algebra, 60 hours/year
- Supervision of projects, 30 hours/year
- Data Valorization, 46 hours/year
- Fundamentals in Analysis I & II, 60 hours/year

Teaching Assistant

Université of Côte d'Azur

Oct. 2022 - July 2025

Nice, France

- Mathematical Analysis, 56 hours/year
- Linear Algebra, 32 hours/year

Instructor

University of Dschang

Mar. 2021 - June 2021

Dschang, Cameroon

- Introduction to Statistics, 15 hours/year

Publications

1. **Penlap Tamagoua, J.**, Baldazzi, V., Grogard, F., and Touzeau, S., Plant tolerance is explained by resource-based plant–nematode interactions, *Mathematical Biosciences*, 2026, doi:[10.1016/j.mbs.2026.109711](https://doi.org/10.1016/j.mbs.2026.109711).
2. **Penlap Tamagoua, J.**, Touzeau, S., Grogard, F., and Baldazzi, V., Unveiling plant tolerance mechanisms through a unified plant–nematode model, 2026, [Preprint](#).
3. Mailleret et al., When seasonality constrains farmer decisions: coupled dynamics of plant disease and clean seed use, 2026, [Preprint](#).
4. Plex Sulá et al., Nine principles for geographic surveillance of biological invasions, 2026, working paper.

Selected Talks & Poster

- 16th European Conference on Mathematical and Theoretical Biology (ECMTB), Graz, Austria (2026). *Plant tolerance is explained by resource-based plant–nematode interactions*.
- 14th European Conference on Mathematical and Theoretical Biology (ECMTB), Toledo, Spain (2024). *Coupling plant physiology and pest demography to understand plant-pest interactions*.
- Plant Pathology Department | Global Food Systems Institute, University of Florida, USA (2023). *Modelling plant–nematode interactions to understand plant tolerance*.
- Mathematical Population Dynamics, Ecology and Evolution (MPDEE), Marseille, France (2023). *Modelling plant–nematode interactions to understand plant tolerance*.
- 13th Dynamical Systems Applied to Biology and Natural Sciences (DSABNS), Bilbao, Spain (2023). Poster. *Modelling plant–nematode interactions to understand plant tolerance*.

Student Supervision

- Del Rosso Luca, Le Potier Killian, Hazo Alexandre, *Démixage de sources musicales par Mixture of Experts*, Final year of Engineering, Polytech Nice Sophia, joint supervision with Lionel Fillatre, 2025-2026 (Projet d'Etudes et de Recherche).
- Lucile Jam, Francesco Orsi, Enzo Rizzo, *Analyse d'étiquettes alimentaires: approches fondées sur les transformers*, Final year of Engineering, Polytech Nice Sophia, joint supervision with Lionel Fillatre, 2025-2026 (Projet d'Etudes et de Recherche).
- Diego Petit, *Clean seeds Model*, 4th Engineering year, Polytech Nice Sophia, joint supervision with Ludovic Mailleret, Frédéric Grogard, Frédéric Hamelin and Israël Tankam, Mai-Sept. 2024 (BEEP project).

Skills

Data & Modelling: Dynamical systems, Optimisation, Sensitivity analysis, Data analysis.

Coding: Python, R, Shell script, GitLab, Streamlit, Gradio.

Operating systems: Linux, Windows.

Cloud computing: Abaca/Grid'5000.

Languages: French (native), English (professional), Ghomà hom (dialect).

Awards and Grants

Inria-INRAE PhD Fellowship , Inria Centre at Université de Côte d'Azur.	Oct. 2022 - Sept. 2025
MasterCard Foundation MSc Scholarship for studies at AIMS.	Sept. 2021 - July 2022
President Paul Biya's Academic Excellence Prize , yearly granted to the best students in Cameroon.	Oct. 2018 - Oct. 2021
Olympiades Universitaires du Conseil Africain et Malgache pour l'Enseignement Supérieur (CAMES), winner of the national selection in mathematics.	Jan. 2020

Miscellaneous

Scientific mediation: <i>"How to use mathematics to solve problems in Agriculture?"</i> to Terra Numerica high school students.	Jan 2023
Hobbies: Football, Climbing, Hiking, Traveling, Reading.	

References

Frédéric Grognaud frederic.grognaud@inria.fr	Valentina Baldazzi valentina.baldazzi@inrae.fr	Suzanne Touzeau suzanne.touzeau@inrae.fr
---	---	---